An All-Period Nonlattice Suspension Determinant with Exact Clock-Sector Separation

Route-A structural certificate C130

Abstract

We freeze the full binary shift with roof values 1 and $\sqrt{2}$. Its two-state bivariate matrix has determinant $1-u-v$; Laplace specialization gives the entire exponential polynomial $1-e^{-s}-e^{-\sqrt{2}s}$. The same source owns an all-period primitive dynamical Euler/trace identity. Irrationality separates symbol-count clock sectors and forbids a nonzero imaginary period, while the rational roof $(1,2)$ restores collisions and vertical periodicity. These are source-dynamical statements, not a target divisor or arithmetic correspondence.

1 Frozen mixing suspension

Let $B = \left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right)$ and let Σ_B be the two-sided full binary shift. Since B is positive, the shift is mixing. Freeze $\tau(0)=1$, $\tau(1)=\sqrt{2}$ and form $X^\tau = (\Sigma_B \times \mathbb{R})/((x, t+\tau(x)) \sim (\sigma x, t))$. For a closed word w, write $N_j(w)$ for its symbol populations and

$$\ell(w) = N_0(w) + \sqrt{2}N_1(w).$$

2 Bivariate owner and primitive identity

Destination-symbol weights give

$$M(u,v) = B\,\mathrm{diag}(u,v) = \begin{pmatrix} u & v \\ u & v \end{pmatrix}, \qquad \Delta(u,v) = \det(I-M) = 1-u-v.$$

The eigenvalues are $u+v, 0$, hence for every $n \geq 1$,

$$\mathrm{Tr}\, M^n = (u+v)^n = \sum_{k=0}^{n} \binom{n}{k} u^{n-k} v^k. \tag{1}$$

The trace sums all rooted binary closed words. For $|u|+|v|<1$,

$$-\log \Delta(u,v) = \sum_{n\geq 1} \frac{\mathrm{Tr}\, M(u,v)^n}{n}.$$

Decomposing each word uniquely as a repetition of a primitive cyclic word γ and regrouping this absolutely convergent series gives

$$\Delta(u,v) = \prod_{[\gamma]} (1 - u^{N_0(\gamma)} v^{N_1(\gamma)}). \tag{2}$$

Thus with $u = e^{-s}$ and $v = e^{-\sqrt{2}s}$,

$$d_\tau(s) = 1 - e^{-s} - e^{-\sqrt{2}s} = \prod_{[\gamma]} (1 - e^{-s\ell(\gamma)}). \tag{3}$$

Equation (2) also holds formally in the total-degree completion, since each degree sees finitely many cycles. Let $h>0$ solve $e^{-h} + e^{-\sqrt{2}h} = 1$. The product (3) converges absolutely for $\mathrm{Re}\, s > h$. The explicit left side is entire and its reciprocal is meromorphic; the product is not asserted to converge globally.

3 Irrational clock and rational control

If $a + b\sqrt{2} = c + d\sqrt{2}$ for integers, irrationality gives $(a,b)=(c,d)$. Thus different population vectors never collide in suspension time. This is not orbit injectivity: the distinct primitive necklaces 000111 and 001011 both have vector $(3,3)$ and roof $3+3\sqrt{2}$.

The same independence forbids a nonzero imaginary period. Indeed, $d_\tau(s+iT)=d_\tau(s)$ for every s forces $e^{-iT}=e^{-i\sqrt{2}T}=1$ by separating the two exponential coefficients. Thus both $T/(2\pi)$ and $\sqrt{2}T/(2\pi)$ are integers, so $T=0$.

For the control, change only the roof to $(1,2)$. Then

$$d_{\mathrm{rat}}(s)=1-e^{-s}-e^{-2s}=1-q-q^2,\qquad q=e^{-s}.$$

The second repetition of [0], with counts $(2,0)$, collides at time 2 with primitive fixed orbit [1], with counts $(0,1)$. Also $d_{\mathrm{rat}}(s+2\pi i)=d_{\mathrm{rat}}(s)$: lattice periodicity is restored.

4 Replay and Route-A boundary

Exact replay gives

n	1	2	3	4	5	6	7	8	9	10
rooted	2	4	8	16	32	64	128	256	512	1024
primitive	2	1	2	3	6	9	18	30	56	99
sectors	2	3	4	5	6	7	8	9	10	11

Totals are 2046 rooted words, 226 primitive cycles, and 65 sectors. The independent standard-library checker (139 assertions), fresh symbolic reconstruction (110 checks), and byte replay pass. The hostile suite rejects all 43 repaired-hash semantic forgeries and one separate stale-hash mutation.

The strict tuple is (`A1_WEAK`, `A2_FAIL`, `A3_FAIL`, `A4_FAIL`), overall `ROUTE_A_EXPLORATORY`; Route B is disabled. We claim no arithmetic Euler factors, root number, target divisor, functional equation, automorphy, natural Hilbert–Pólya lift, or orbit-level injectivity within a sector. The literal firewall is `NO_BAD_EULER_OR_ROOT_NUMBER`.